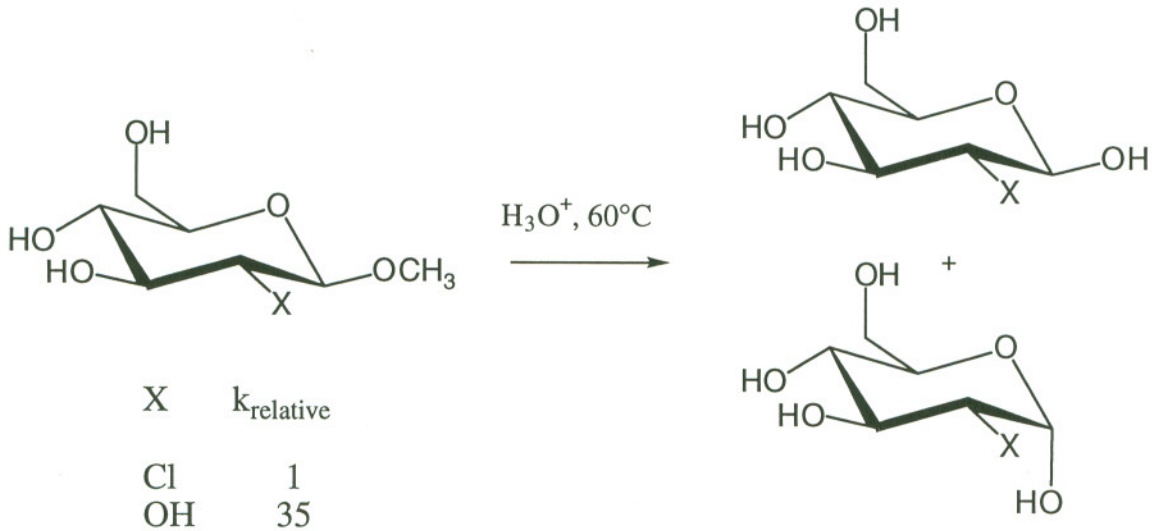
Organic Division Cumulative Exam October 8, 1999

Answer **four** of the five questions below.

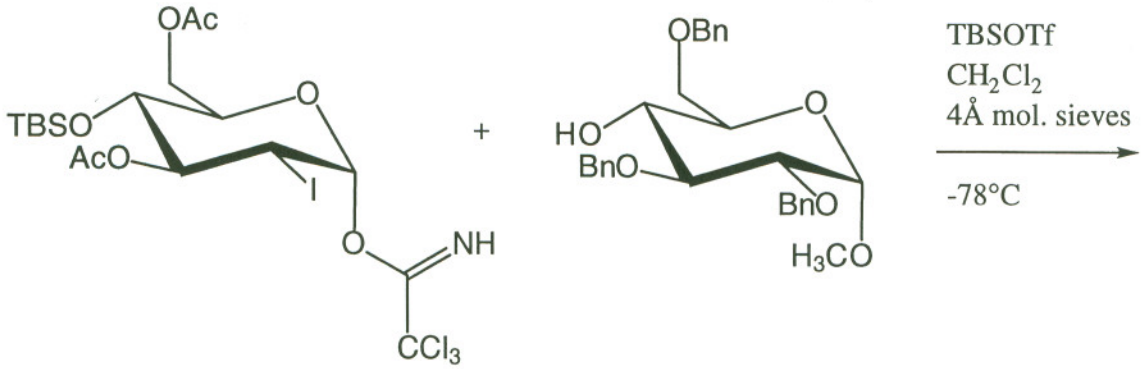
(1) Rationalize the following series of relative rates of acid hydrolysis of the indicated sugars.



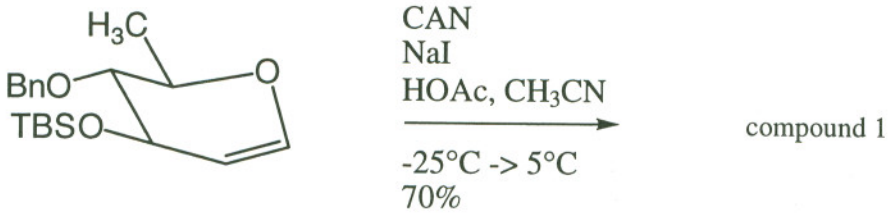
X	$k_{relative}$
Cl	1
OH	35
H	104

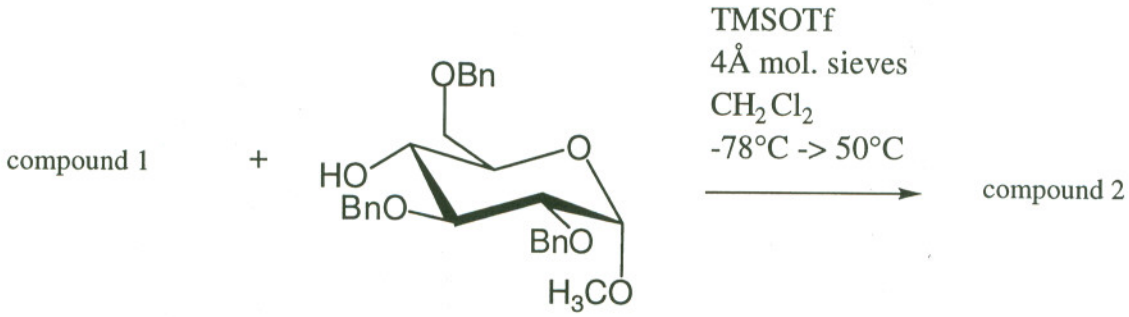
(2) Given the answer to the above question, what is one advantage of Roush's procedure for making the oligosaccharide components of complex organic molecules by reduction of 2-iodo- and 2-bromosugars as one the final steps rather than attaching the 2-deoxy sugar at an early stage of the synthesis?

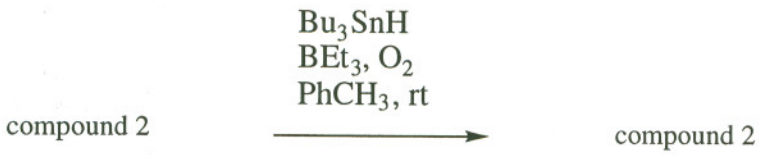
(3) Draw the product of the following reaction. Be sure to draw the products so as to unambiguously indicate stereochemistry.



(4) Draw the structures of compounds 1, 2 and 3.







(5) Briefly discuss one significant point of Roush's seminar not covered in the above questions.